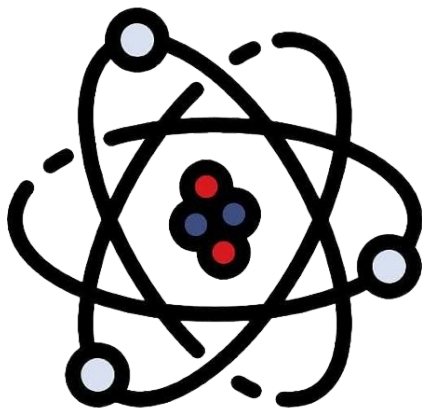


A STEAMLabs x Simistry Project

PHYSICS

EXPERIMENTS



HANDBOOK

Introduction

Welcome to the **PhysiXplorers Handbook**! At PhysiXplorers, our mission is to bring hands-on physics education to students from all backgrounds and help them explore the wonders of the universe. This kit is designed to cover 5 major areas of physics with a hands-on experiment corresponding to each concept, from mechanics & waves to electricity and magnetism. Each activity is designed to help you learn physics by building and testing it in action. This handbook includes step by step instructions, explanations of the physics behind each experiment, and interactive online simulations (made in partnership with Simistry) to help you explore the concept further—from the comfort of your bedroom! You can complete the experiments in any order, but they vary in difficulty. To deepen your understanding, complete the science report at the end of each activity!

If you are new to physics/engineering labs, we recommend the following progression:

1. Straw Rocket
2. Periscope
3. Straw Pan Flute
4. Circuit Art
5. Electric Motor

Free, printable handouts & templates for writing a science report are available on our website! Scan the QR code at the end of the packet, or visit PhysiXploreLab.org for these resources. Let's embark on the miraculous journey of physics by exploring the mysteries of the universe!

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What's Inside Your Kit?

Straw Rocket

- 1 Straw
- 1 x long rectangular piece of paper
- 1 x short rectangular piece of paper
- 2 x Cardstock Paper
- Tape

Periscope

- 2 x 1-inch Mirror Stickers
- 1 x Periscope Template
- 1 x Cardstock Paper

Straw Pan Flute

- 8 x Straw
- 1 x Paper Ruler
- Tape

Circuit Art

- 3 x LED light bulbs
- 1 x 8 inch strip copper tape
- 1 x 10 inch strip copper tape
- 1 x printer paper
- 1 x button battery

Electric Motor

- 1 x AAA Battery
- 4 x Magnets
- 2 x Paperclips
- 12 in. Copper Wire
- Sandpaper
- Masking Tape

Getting Started...

For the best experience, please have scissors & tape available. All other materials are included in the kit!

Straw Rocket Instructions

Explore the Science



Scan the QR code to explore an interactive simulation and see how thrust, air resistance and gravity work before you begin.

Materials:

- 1 Straw
- 1 x long rectangular piece of paper
- 1 x short rectangular piece of paper
- 2 x Cardstock Paper
- Tape

Procedures

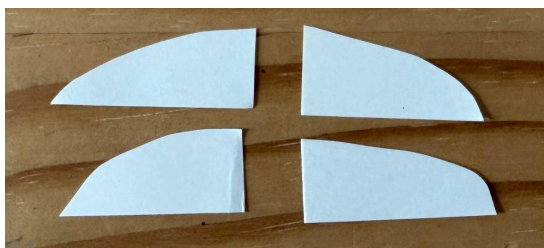
1. Roll up one of the rectangular pieces of paper (can either be the longer or shorter one)



2. Flatten one end of the rocket body, then fold it so that it makes one pointed edge. Stabilize it by using tape



3. Now, cut out the fins from the cardstock paper & tape to the rocket! You can make it any shape you like!



PhysiXplore Lab Science Report

Activity: _____

Name: _____ Date: _____

Partner(s): _____

Methods

Describe what you did during the activity!

Draw Your Experiment!

Results

What did you notice? What happened when you changed something?

Conclusion

What (if anything) would you change? What did you discover/learn?

Periscope Instructions

Explore the Science



Scan the QR code to explore an interactive simulation and see how mirrors, light, and optics work before you begin!

Materials:

- 2 x 1-inch Mirror Stickers
- 1 x Periscope Template
- 1 x Cardstock Paper

Procedures

1. Put the template paper on top of the cardstock (hard) paper. Then, cut along the **solid lines** of the template & cardstock paper together. *Do not cut the dotted lines!*
2. Keeping the template paper on top of the hard (cardstock) paper, fold along the dotted lines of the template & cardstock paper together
3. Now, you can remove the template paper (you won't need it again)! Fold & tape the cardstock so that it forms the shape of the periscope:



4. Remove the white paper from the back of the mirror tile, and stick it onto the end of the periscope:



5. Once the stickers are in the correct place, **remove the thin film** from the front of the sticker! *If you don't do this, your periscope **won't** work*

PhysiXplore Lab Science Report

Activity: _____

Name: _____ Date: _____

Partner(s): _____

Methods

Describe what you did during the activity!

Draw Your Experiment!

Results

What did you notice? What happened when you changed something?

Conclusion

What (if anything) would you change? What did you discover/learn?

Straw Pan Flute Instructions

Explore the Science



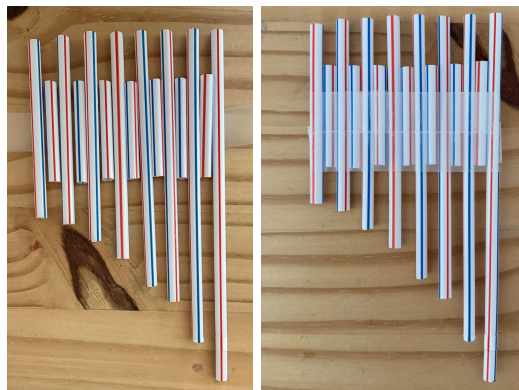
Scan the QR code to explore an interactive simulation and see how waves & sound are related before you begin, as well as the different properties of sound!

Materials:

- 8 x Straw
- 1 x Paper Ruler
- Tape

Procedures

1. Cut the straws to the following lengths. Save the extra pieces of straw!
 - Straw 1 = 17.5 cm
 - Straw 2 = 15.5 cm
 - Straw 3 = 13.5 cm
 - Straw 4 = 12.5 cm
 - Straw 5 = 11 cm
 - Straw 6 = 10 cm
 - Straw 7 = 9 cm
 - Straw 8 = 8 cm
2. Tape the main straw lengths & the extra pieces of straws alternately, in the pattern below:



3. You are done! Blow through the top of the flute, and explore how the sound changes as the straws get shorter. What happens when you blow through a straw with one end blocked

PhysiXplore Lab Science Report

Activity: _____

Name: _____ Date: _____

Partner(s): _____

Methods

Describe what you did during the activity!

Draw Your Experiment!

Results

What did you notice? What happened when you changed something?

Conclusion

What (if anything) would you change? What did you discover/learn?

Circuit Art Instructions

Explore the Science



Scan the QR code to explore an interactive simulation and see how circuits, current, and LEDs work before you begin.

Materials:

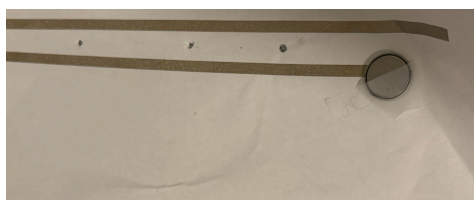
- 3 x LED light bulbs
- 1 x 8 inch strip copper tape
- 1 x 10 inch strip copper tape
- 1 x printer paper
- 1 x button battery

Procedures

1. Put the two pieces of tape onto the printer paper in the arrangement below. Then, place the battery so that the positive side (labeled "+") faces down & touches the shorter copper tape.

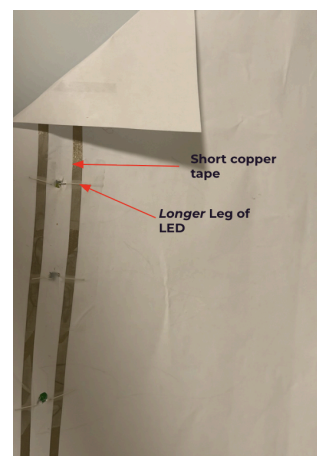


2. Tape the battery onto the printer paper. Make sure to leave at least $\frac{1}{2}$ of the battery exposed so that the longer piece of tape can directly touch the rough "-" side of the battery. Example below:
3. Poke 3 holes (at roughly equal distance to both of the copper tape) for the places where you want to place the LED lights:



If you have trouble with this step, ask an adult for help! A sharp pencil is recommended

4. Look closely at the LED light. One side has a longer leg while the other leg is shorter. The longer leg is the positive "+" terminal. Orient the light bulb so that the **longer leg** of the LED light touches the **shorter piece of copper tape** (the one the battery is taped to). After this, tape the light bulbs tightly to the copper tape—make sure the connection is secure:



Pro Tip: Before taping, make sure your light bulb lights up after putting it into the correct orientation!

5. Fold the paper so that the longer end of the copper tape touches the negative "-" terminal of the button battery. Your design should light up now!

PhysiXplore Lab Science Report

Activity: _____

Name: _____ Date: _____

Partner(s): _____

Methods

Describe what you did during the activity!

Draw Your Experiment!

Results

What did you notice? What happened when you changed something?

Conclusion

What (if anything) would you change? What did you discover/learn?

Electric Motor Instructions

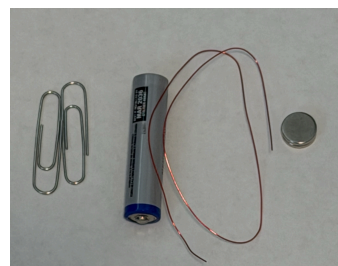
Explore the Science



Scan the QR code to explore an interactive simulation and see how electromagnetism works together to power an electric motor before you begin!

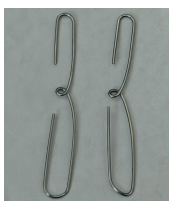
Materials:

- 1 x AAA Battery
- 4 x Magnets
- 2 x Paperclips
- 12 in. Copper Wire
- Sandpaper
- Masking Tape

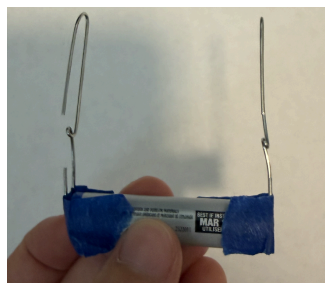


Procedures

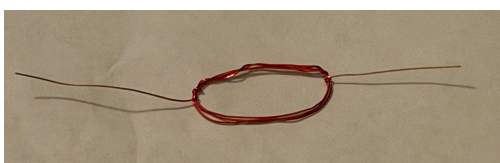
1. Bend the paper clips (as shown below). Make the loops as tight as you can



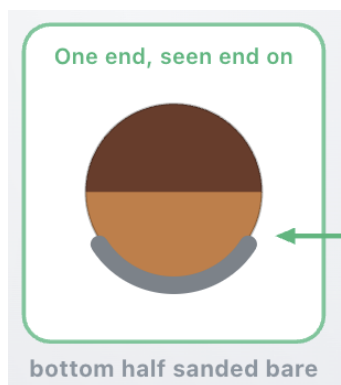
2. Attach these to the battery. Make sure it has a good connection with the terminal:



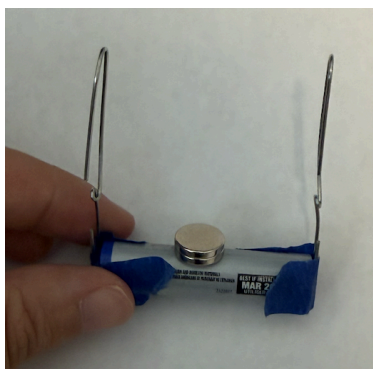
3. Make a ring with the copper wire. Start at the middle of the wire and wrap it around your fingers to form a ring. Then, twist both ends of the wire into coils around the ring, as shown in the diagram below. Try to wrap the coil as tightly as possible:



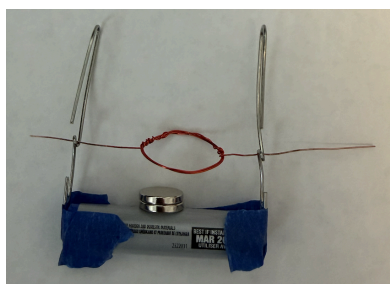
4. Using sandpaper to remove the red copper coating for half of the wire on each side:



5. Attach the magnets to the battery



6. Put the loop through the paper clips. It might take a bit of adjustments to get it working! **Don't leave the wire on for too long—it will get hot!**



If it doesn't work, here's some things to check:

- Is your wire hot? If not, try sanding down more of the wire loop & make sure your connections are good!
- Is your wire sagging? Try adjusting the shape/how many times you wrap the wire to decrease the weight

PhysiXplore Lab Science Report

Activity: _____

Name: _____ Date: _____

Partner(s): _____

Methods

Describe what you did during the activity!

Draw Your Experiment!

Results

What did you notice? What happened when you changed something?

Conclusion

What (if anything) would you change? What did you discover/learn?

Conclusion

Thank you for exploring all areas of physics with us through the PhysiXplorer Handbook! We hope these 5 activities have sparked your curiosity and a deeper passion for understanding the world through physics.

But your exploration doesn't have to end here! Keep asking questions, testing ideas, and looking for physics in the world around you—there are plenty of examples we haven't covered in this kit! Keep exploring, experimenting, and wondering!

Physics is everywhere—you just have to look for it.

Keep Exploring with PhysiXplore Lab

Want to see what's next? Scan the QR code below to visit our website for new projects, activities, and updates. Our mission is to make physics hands-on, accessible, and exciting for students everywhere. As PhysiXplorer grows, we're working to reach more students and create new ways for young learners to explore the world through physics. All proceeds from PhysiXplorer kits support physics education through the American Physical Society (APS), helping expand opportunities for K–12 students to discover and engage with physics.